

Tree

Definition: A *tree* is a connected undirected graph with no simple circuits.

Or

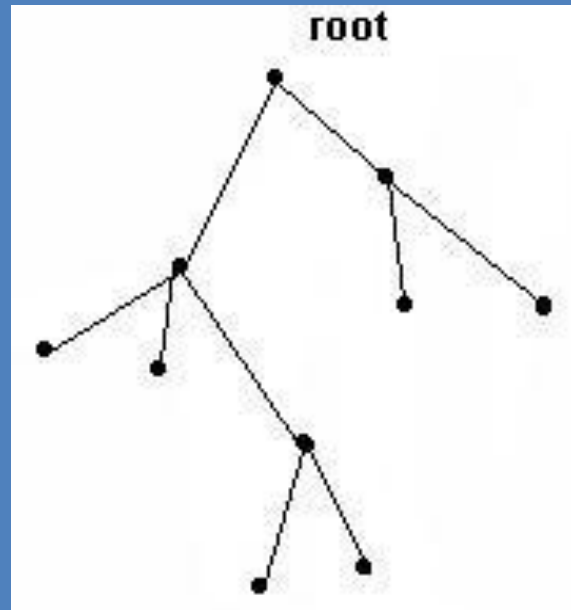
A *tree* T is

- A simple graph such that for every pair of vertices v and w , there is a unique path from v to w .



Rooted tree

A *rooted tree* is a tree where one of its vertices is designated the *root*



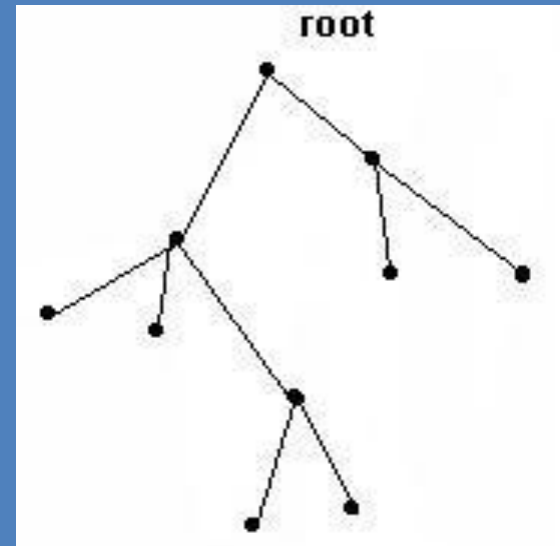
Level of a vertex and tree height

Let T be a rooted tree:

- The *level* $l(v)$ of a vertex v is the length of the simple path from v to the root of the tree
- The *height* h of a rooted tree T is the maximum of all level numbers of its vertices:

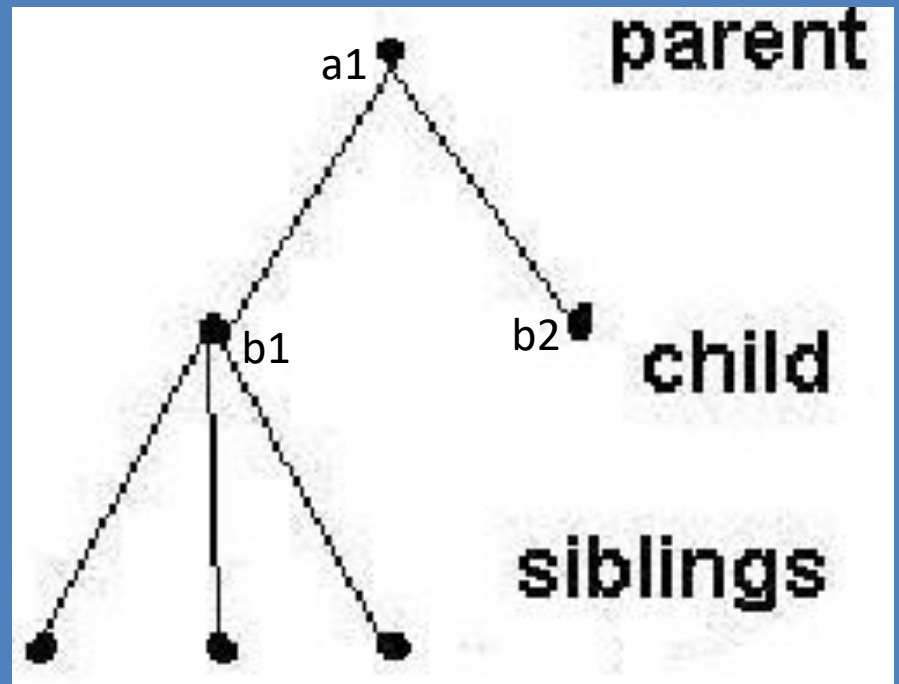
$$h = \max_{v \in V(T)} \{ l(v) \}$$

- Example:
 - the tree on the right has height 3



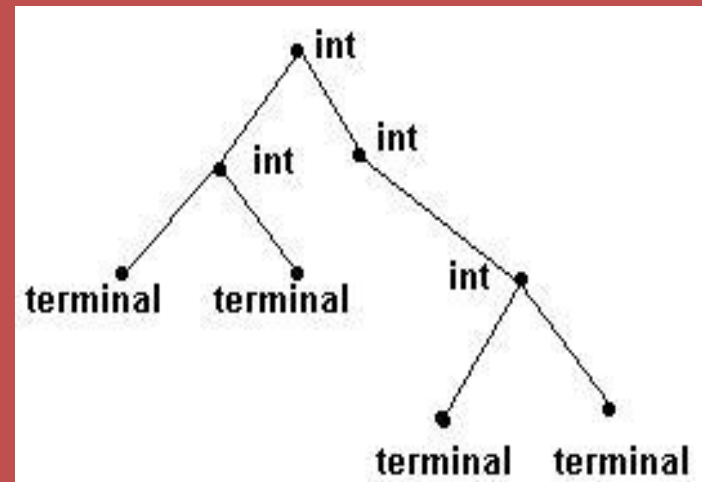
Terminology

- Parent
- Ancestor
- Child
- Descendant
- Siblings
- Terminal vertices
- Internal vertices
- Subtrees



Internal and external vertices

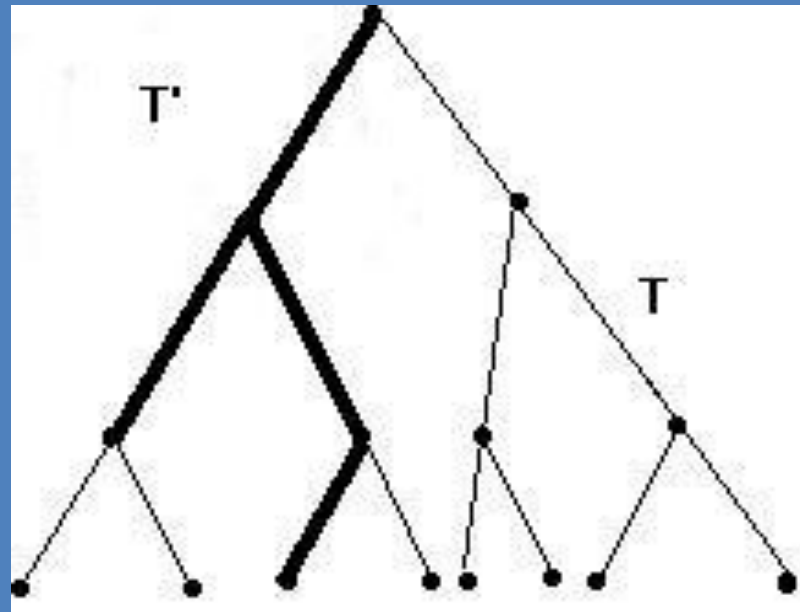
- An *internal vertex* is a vertex that has at least one child
- A *terminal vertex* is a vertex that has no children
- The tree in the example has 4 internal vertices and 4 terminal vertices



Subtrees

A subtree of a tree T is a tree T' such that

- $V(T') \subseteq V(T)$ and
- $E(T') \subseteq E(T)$



Characterization of trees

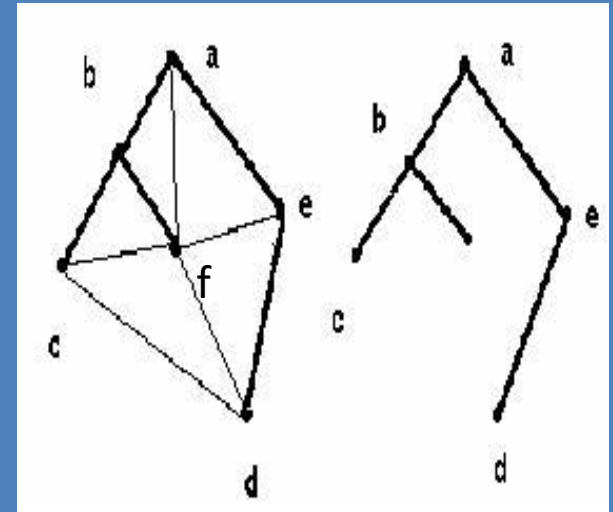
If T is a graph with n vertices, the following are equivalent:

- a) T is a tree
- b) T is connected and acyclic
 - (“acyclic” = having no cycles)
- c) T is connected and has $n-1$ edges
- d) T is acyclic and has $n-1$ edges

Spanning trees

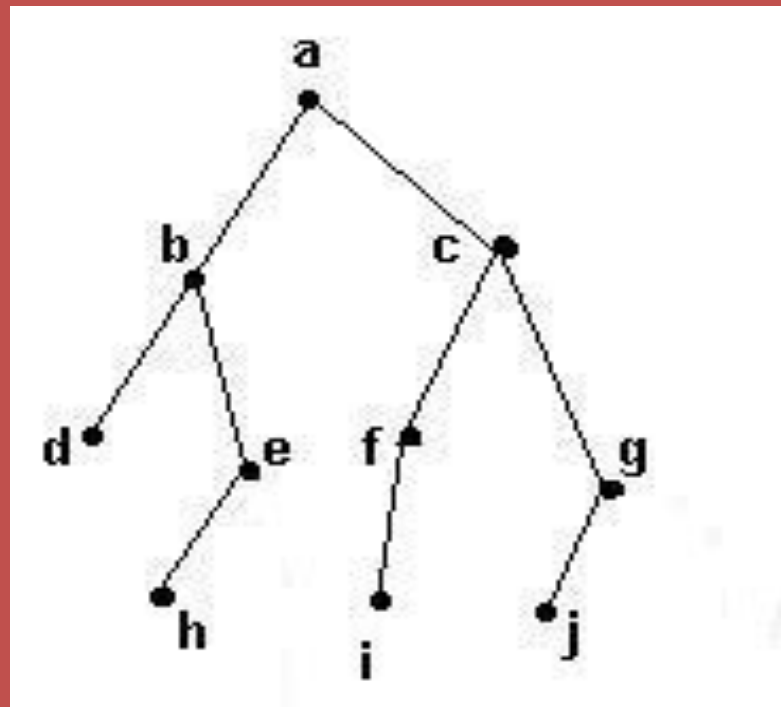
Given a graph G , a tree T is a *spanning tree* of G if:

- T is a subgraph of G
and
- T contains all the vertices of G



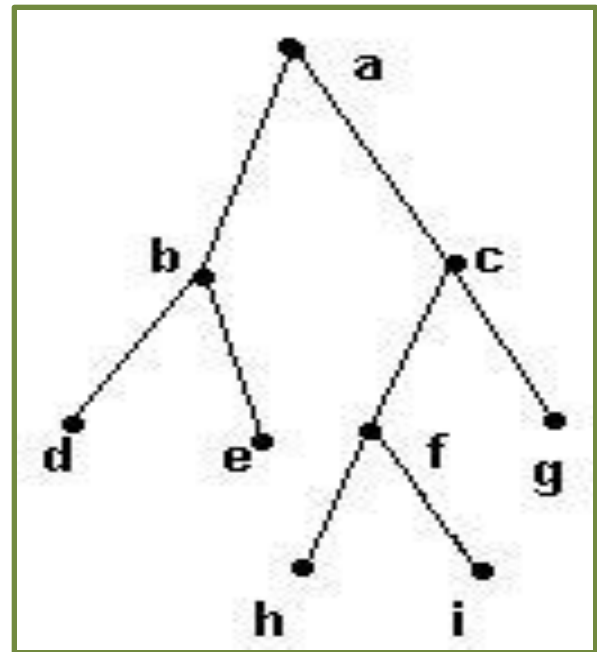
Binary trees

A *binary tree* is a tree where each vertex has zero, one or two children



Full binary tree

A *full* binary tree is a binary tree in which each vertex has two or no children.



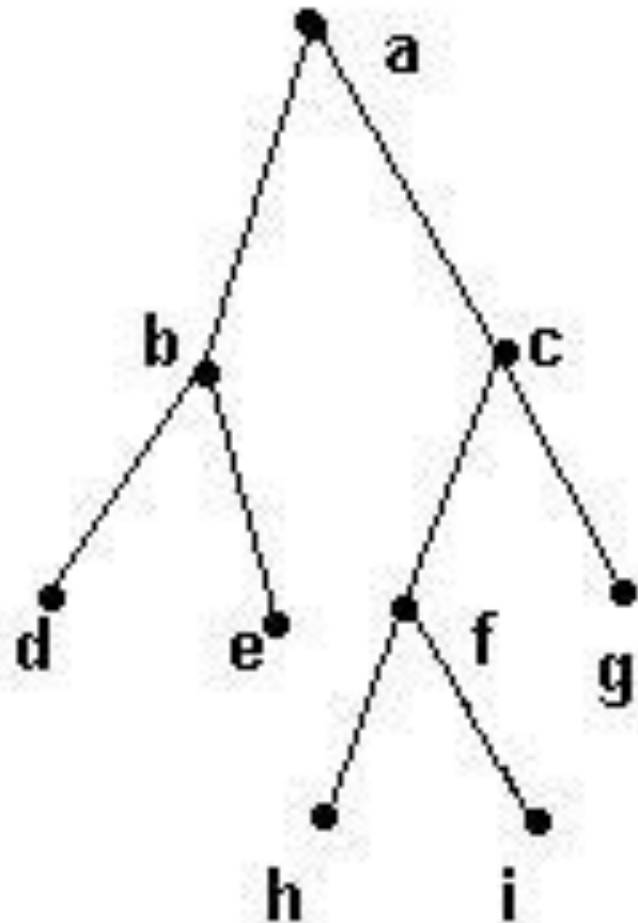
Full binary tree

Theorem 7.5.4: If T is a *full binary tree* with k internal vertices, then

□ T has $k + 1$ terminal vertices and

□ the total number of vertices is $2k + 1$.

- Example: there are $k = 4$ internal vertices (a, b, c and f) and 5 terminal vertices (d, e, g, h and i) for a total of 9 vertices.

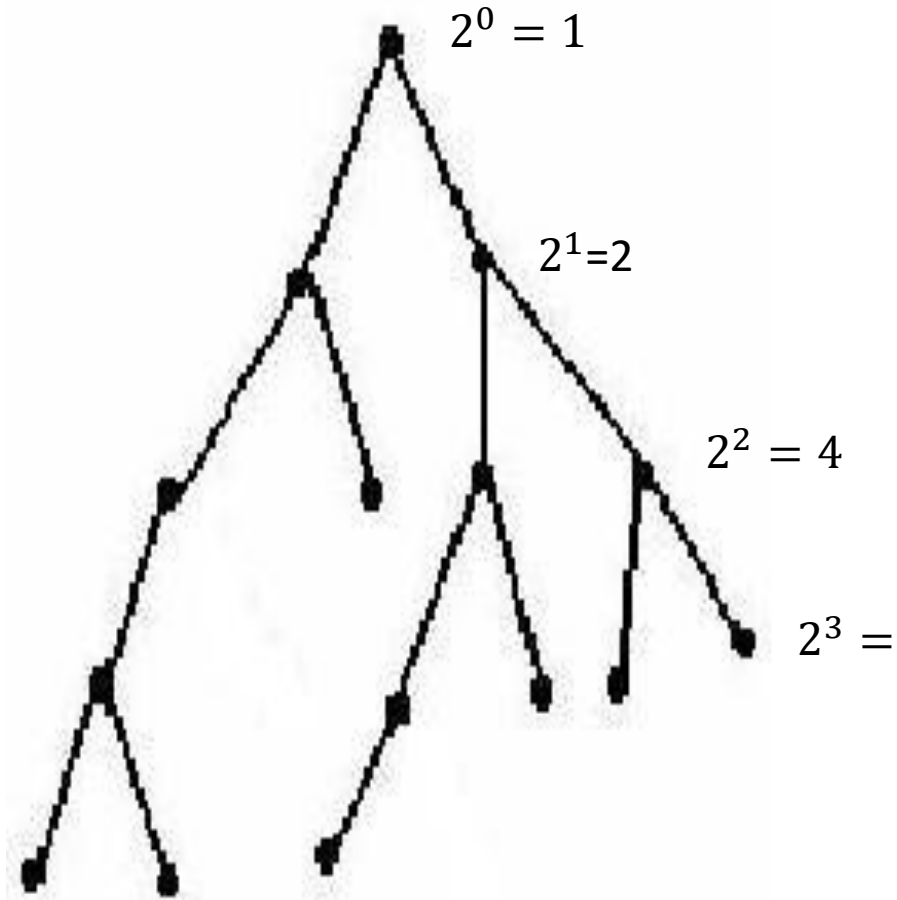


Height and terminal vertices

- Theorem 7.5.6: If a binary tree of height h has t terminal vertices, then $\log_2 t \leq h$, where $\log$ is logarithm base 2.

Equivalently, $t \leq 2^h$.

- Example, $h = 4$ and $t = 7$. Then: $t = 7 < 16 = 2^4 = 2^h$



A case of equality

- If all t terminal vertices of a full binary tree T have the same level $h =$ height of T , then
$$t = 2^h.$$
- Example:
 - The height is $h = 3$,
 - and the number of terminal vertices is $t = 8$
 - $t = 8 = 2^3 = 2^h$

